Ans 3. Photoelectric Effect: Emission of electrons from metal surface when light of suitable frequency hits.

Einstein Equation: E_photon = W0 + KE_max where W0 is work function of the metal.

Ans 1. Newton 2nd Law states rate of change of momentum is directly proportional to applied force.

F = dp/dt = d(mv)/dt = m(dv/dt) = ma.

Ans 11(a) Ohm's Law: V = IR at constant temperature.

Resistance factors: R = rho * L / A.

Ans 11(b) Parallel resistors: Total current I = I1 + I2.

V/Req = V/R1 + V/R2 => 1/Req = 1/R1 + 1/R2.

Ans 5. Transformer working principle: Mutual Induction between primary and secondary coils.

Turns ratio relation: Vs / Vp = Ns / Np. [Continued on Page 2]

Ans 5 (Cont.)

1. Copper Loss: $I^2 R$ heating in primary & secondary turns.
2. Eddy Current Loss: Induced circulating currents (laminated core mitigates).
3. Hysteresis Loss: Reversal of magnetisation in core.

Ans 2.

m = 4 kg, v = 10 m/s. Kinetic Energy KE = 1/2 m v^2 = 0.5 * 4 * 100 = 200 Joules.

Extra Note

Rough note on Maxwell equations: del.E = rho/epsilon0, del.B = 0.